Test Data

$z_1 = \begin{pmatrix} 30 \\ 15 \end{pmatrix}$ y=60 $z_2 = \begin{pmatrix} 40 \\ 20 \end{pmatrix}$ y=80 $z_3 = \begin{pmatrix} 50 \\ 25 \end{pmatrix}$ y=100

Outlier X_s	NO Regularizer with (as if λ=0)	Ridge Regularizer with λ=3	Ridge Regularizer with λ=10
$\begin{pmatrix} 11 \\ 3 \end{pmatrix}$ 22	W = (0, 2, 0), ‖W‖²=4	W = (0.1, 1.95, 0.03), ‖W‖²=3.8	W = (0.13, 1.87, 0.09), ‖W‖²=3.5
$\begin{pmatrix} 30 \\ 3 \end{pmatrix}$ 22	W = (2.7, 0.67, 0.64), ‖W‖²=8.11	W = (1.2, 0.72, 0.77), ‖W‖²=2.45	W = (0.54, 0.73, 0.81), ‖W‖²=1.48
$\begin{pmatrix} 60 \\ 3 \end{pmatrix}$ 22	W = (4, 0.29, 0.78), ‖W‖²=16.7	W = (1.77, 0.32, 0.99), ‖W‖²=4.2	W = (0.82, 0.33, 1.1), ‖W‖²=1.92

NO Regularizer with (as if λ=0)

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	60	80	100

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	32.4	42.3	52.3

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	24.4	31.2	38

Ridge Regularizer with λ=3

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	59	78.7	98.4

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	34.1	45.1	56.1

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	26.2	34.4	42.5

Ridge Regularizer with λ=10

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	57.4	76.5	95.6

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	34.6	46	57.4

	Test₁	Test₂	Test₃
x₁	30	40	50
x₂	15	20	25
y	26.8	35.4	44.1